

VOLUMETRIC ANALYSIS

AIM: Preparation of 100 ml of 0.05 M solution of Mohr's salt.

THEORY:

Molecular mass of Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O} = 392$

Hence, for preparing 1000 ml of 1 M Mohr's salt solution, Mohr's salt required = 392 g

For preparing 100 ml of M/20 Mohr's salt solution,

$$\text{mass} = \frac{\text{Molarity} \times \text{molar mass} \times \text{volume}}{1000}$$

$$\text{Mass of Mohr's salt required} = \frac{0.05 \times 392 \times 100}{1000} = 1.96\text{g}$$

APPARATUS:

Watch glass, weight box, fractional weight box, 100 ml beaker, glass rod, 100 ml standard flask, dropper and wash bottle.

CHEMICALS REQUIRED:

Mohr's salt, conc. H_2SO_4 and distilled water.

PROCEDURE:

1. Weigh the clean and dry watch glass and record its weight in the notebook.
2. Weigh accurately 1.96 g of Mohr's salt crystals on the watch glass and record the weight in the notebook.
3. Transfer carefully the weighed Mohr's salt from the watch glass into a clean 100 ml measuring flask using a funnel.
4. Wash the watch glass thoroughly with distilled water to transfer the sticking salt completely into the flask.
Dissolve the salt in the standard flask with gentle stirring.
5. Add about 1 test tube of dilute. H_2SO_4 to the solution in the standard flask to prevent the hydrolysis of

ferrous sulphate.

6. Wash the funnel with distilled water and transfer the washings into the measuring flask.

7. Add enough distilled water to the standard flask carefully upto just below the etched mark on its neck with the help of wash bottle.

8. Add the last few drops of distilled water with a pipette or a dropper until the lower level of the meniscus just touches the mark on the standard flask.

9. Close the standard flask with a stopper and shake it gently to make the solution homogeneous (i.e., uniform throughout) and label it as M/20 Mohr's salt solution.

FREEDOM INTERNATIONAL SCHOOL

VOLUMETRIC ANALYSIS

AIM: Prepare 0.05 M solution of ferrous ammonium sulphate (Mohr's salt). Using this solution find out the molarity and strength of the given KMnO_4 solution.

INDICATOR: KMnO_4 is a self-indicator.

END POINT: Colourless to permanent pale pink colour.

PROCEDURE:

1. Prepare 100 ml of 0.05 M Mohr's salt solution by dissolving 1.96 g of Mohr's salt in water

Rinse the pipette with the 0.05 M Mohr's salt solution and pipette out 10 ml of it in a washed titration flask.

2. Rinse and fill the burette with the given KMnO_4 solution.

3. Add one test tube full of dilute sulphuric acid ($\sim 2\text{ M}$) to the solution in the titration flask.

4. Note the initial reading of the burette.

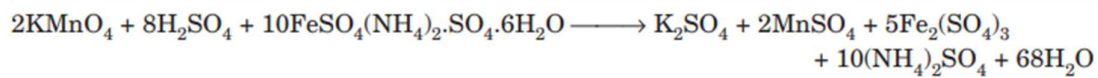
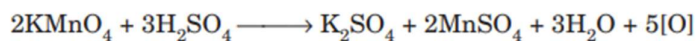
5. Now add KMnO_4 solution from the burette till a permanent pale pink colour is imparted to the solution in the titration flask on addition of the last single drop of KMnO_4 solution.

6. Note the final reading of the burette.

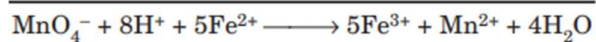
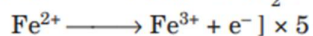
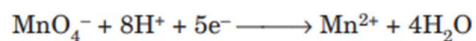
7. Repeat the above steps 3–4 times to get a set of three concordant readings.

CHEMICAL EQUATIONS

Molecular equations



Ionic equations



RESULT:

1. The molarity of the given KMnO_4 solution is = _____ M

2. The strength of the given KMnO_4 solution is = _____ g/L

FREEDOM INTERNATIONAL SCHOOL

OBSERVATION:

Weight of Mohr's salt = 1.96 g

Volume of Mohr's salt solution prepared = 100 ml

Molarity of Mohr's salt solution = 0.05 M

Volume of Mohr's salt solution taken for each titration = 10 ml

S. No.	Burette Readings		Volume of the KMnO ₄ solution used
	Initial	Final	
1.			
2.			
3.			
4.			

Concordant volume = _____ ml

CALCULATIONS:

(a) Molarity of the KMnO₄ solution:

$$\frac{M_{\text{KMnO}_4} \times V_{\text{KMnO}_4}}{M_{\text{Mohr's salt}} \times V_{\text{Mohr's salt}}} = \frac{2}{10}$$

$$\frac{M_{\text{KMnO}_4} \times 10}{0.05 \times 10} = \frac{2}{10}$$

$$M_{\text{KMnO}_4} = \text{_____ mol/L}$$

(b) Strength of the KMnO₄ solution:

$$\text{Strength (in g/L)} = \text{Molarity} \times \text{Molar mass}$$

$$= \text{_____} \times 158$$

$$= \text{_____ g/L}$$

VOLUMETRIC ANALYSIS

AIM: Preparation of 100 ml of M/40 solution of oxalic acid.

THEORY:

Molecular mass of oxalic acid, $\left(\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array} \cdot 2\text{H}_2\text{O} \right) = 126$

Hence, for preparing 1000 ml of 1M oxalic acid, weight of oxalic acid crystals required = 126 g

For preparing 100 ml of M/40 oxalic acid solution,

$$\text{mass} = \frac{\text{Molarity} \times \text{molar mass} \times \text{volume}}{1000}$$

$$\text{Mass of oxalic acid required} = \frac{0.025 \times 126 \times 100}{1000} = 0.315\text{g}$$

APPARATUS:

Watch glass, weight box, fractional weight box, 100 ml beaker, glass rod, 100 ml standard flask, dropper and wash bottle.

CHEMICALS REQUIRED:

Oxalic acid crystals and distilled water.

PROCEDURE:

1. Weigh the clean and dry watch glass and record its weight in the notebook.
2. Weigh accurately 0.315 g of oxalic acid crystals on the watch glass and record the weight in the notebook.
3. Transfer carefully the weighed oxalic acid from the watch glass into a clean 100 ml standard flask using a funnel.
4. Wash the watch glass thoroughly with distilled water to transfer the sticking salt completely into the flask.
Dissolve the salt in the standard flask with gentle stirring.
5. Finally wash the funnel well with distilled water with the help of a wash bottle to transfer the solution

sticking to the funnel into the standard flask.

6. Swirl the measuring flask till solid oxalic acid dissolves.

7. Add enough distilled water to the standard flask carefully upto just below the etched mark on its neck with the help of wash bottle.

8. Add the last few drops of distilled water with a pipette or a dropper until the lower level of the meniscus just touches the mark on the standard flask.

9. Close the standard flask with a stopper and shake it gently to make the solution homogeneous (i.e., uniform throughout) and label it as M/40 oxalic acid solution.

FREEDOM INTERNATIONAL SCHOOL

VOLUMETRIC ANALYSIS

AIM: Prepare M/40 solution of oxalic acid. Using this solution find out the molarity and strength of the given KMnO_4 solution.

INDICATOR: KMnO_4 is a self-indicator

END POINT: Colourless to permanent pale pink colour

PROCEDURE:

1. Prepare 100 ml of M/40 oxalic acid solution by dissolving 0.315 g of oxalic acid crystals in water

Rinse the pipette with the M/40 oxalic acid solution and pipette out 10 ml of it in a washed titration flask.

2. Rinse and fill the burette with the given KMnO_4 solution.

3. Add one test tube full of dilute sulphuric acid ($\sim 2\text{ M}$) to the solution in titration flask.

4. Note the initial reading of the burette.

5. Heat the flask to $60\text{--}70^\circ\text{C}$ and add KMnO_4 solution from the burette till a permanent pale pink colour is imparted to the solution in the titration flask on addition of a last single drop of KMnO_4 solution

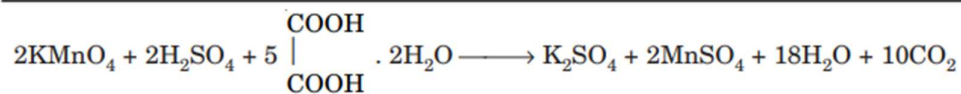
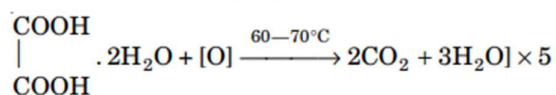
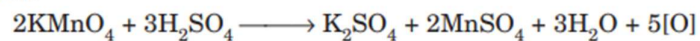
6. Now add KMnO_4 solution from the burette till a permanent pale pink colour is imparted to the solution in the titration flask on addition of last single drop of KMnO_4 solution.

7. Note the final reading of the burette.

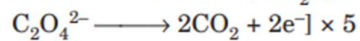
8. Repeat the above steps 3–4 times to get a set of three concordant readings.

CHEMICAL EQUATIONS

Molecular equations



Ionic equations



RESULT:

1. The molarity of the given KMnO_4 solution is = _____ M

2. The strength of the given KMnO_4 solution is = _____ g/L

FREEDOM INTERNATIONAL SCHOOL

OBSERVATION:

Weight of oxalic acid = 0.315 g

Volume of oxalic acid solution prepared = 100 ml

Molarity of oxalic acid solution = M/40

Volume of oxalic acid solution taken for each titration = 10 ml

S. No.	Burette Readings		Volume of the KMnO ₄ solution used
	Initial	Final	
1.			
2.			
3.			
4.			

Concordant volume = _____ ml

CALCULATIONS:

(a) Molarity of the KMnO₄ solution:

$$\frac{M_{\text{KMnO}_4} \times V_{\text{KMnO}_4}}{M_{\text{oxalic acid}} \times V_{\text{oxalic acid}}} = \frac{2}{5}$$

$$M_{\text{KMnO}_4} = \text{_____ mol/L}$$

(b) Strength of the KMnO₄ solution:

$$\text{Strength (in g/L)} = \text{Molarity} \times \text{Molar mass}$$

$$= \text{_____} \times 158$$

$$= \text{_____ g/L}$$